is particularly important. It indicates that these corpuscles, in addition to their function as oxygen-carriers, play an important rôle in metabolism in storing intermediary bodies.

Investigation has been greatly aided by a knowledge of the receptors, since they can be used in the analysis of animal and vegetable poisons to separate them into partial-toxins having different haptophore groups. Thus it is possible by specific locking with specific antibodies to analyze snake poison into neurotoxic, endotheliotoxic and two hemotoxic components. Of the latter, one agglutinates, and the other, through the formation of lecithid, hemolyses the blood (Meyer, Flexner, Kyes). Similar analyses of tetanotoxin and diphtheria toxin are possible. The method is a great advance in a most difficult field of research.

CONCLUSION AND PRACTICAL FORECAST.

The immunization industry which has for its aim the production of sera of highest potency, has made marked progress. Sera of much higher antitoxic value than have hitherto been obtained will be the result. The knowledge that the immunity reaction is reduced by too highly toxic treatment of the animals is spreading. An effort is being made to restrict this immune reaction to tissues of slight vital importance (connective tissue) and to protect vital organs from injury. In this direction are the methods of immunization with toxin and antitoxin injected separately or after admixture. Thus an apparently neutralized toxin may cause antitoxin production (Park). This is due, as will be explained in the next lecture, to the presence of ultra-toxoids.

Progress in the production of curative sera is to be expected only when it becomes possible to obtain antitoxins of greater avidity than those now procured. That this is theoretically possible is shown by the investigations of Kyes who demonstrated that through the immunization of rabbits with cobra-lecithid it is possible to obtain an antibody of considerably higher avidity than that obtained by Calmette with cobra-venom in horses (antivenin). The selection of particularly favorable animal species and the use of toxins of least avidity or ultra-toxones will, most probably, bring this problem much nearer its final solution.

LECTURE II.

PHYSICAL CHEMISTRY *V.* BIOLOGY IN THE DOCTRINES OF IMMUNITY.

Abstract.

INTRODUCTION.

PHYSICO-CHEMICAL investigations of Arrhenius and Madsen concerning the application of physical chemistry to the study of toxins and antitoxins. Their belief that tetanolysin is a uniform substance possessing only a weak affinity to its antitoxin and that its neutralization occurs in accordance with the law of Guldberg and Waage. Attack upon my doctrine of plurality of toxic constituents. Favorable reception of these opposing views in certain scientific circles lacking special competence in sero-therapeutical questions. Protest against the attempt of these authors to claim exclusive priority in the development of these questions. The path of biological investigation — the method of partial neutralization of toxins — followed by them was first opened and by long years of work developed by me. The contention that my investigations represent merely an application of Thomsen's principles in his study of affinities between strong acids and bases is incorrect.

Equally unwarranted is the reproach made by these authors that my investigations are inaccurate and have been conducted without sufficient consideration of sources of error. The most important of my results have been confirmed by them, especially as regards the existence of toxoids and their occurrence with varying degrees of avidity, as "prototoxoids," "syntoxoids." On the contrary, the researches of myself and collaborators have been conducted with the utmost precision and care without consideration of cost or time. Nor can I admit that I have violated the principle of choosing the simplest possible explanations. Such explanations received always my first consideration and, as in the case of diphtheria-toxin, gave place to more complex ones only when the experimental results made this necessary.

POINTS OF DEPARTURE. METHODS AND RESULTS OF PARTIAL NEUTRALIZATION.

It was to be determined whether in the production of toxoids a change in avidity for antitoxin occurred. The only available method for this was that of partial saturation or neutralization. A prerequisite of this method is the exact determination of the test-constants, viz., the simple fatal dose, and the Lo and the L† doses, which have reference to the immunity unit (I. E.), and each of which may be the starting point of the investigation. It is then determined by how many fatal doses the toxicity of a given quantity of toxin, as for example the Lo dose, is lessened by its mixture with a fraction, say one tenth, of the immunity unit of antitoxin. Then with another mixture containing a larger amount of antitoxin a second determination is made, and so on. If the toxin were a single substance of uniform composition and the antitoxin possessed for it a strong affinity, like that between hydrochloric acid and potassium, the quantities of toxin which disappear or are neutralized in these mixtures would be in direct ratio to the quantities of added antitoxin. In fact, however, the process of neutralization takes places after a quite different scheme, as is shown by a glance at the exhibited charts.

It is advantageous to present these experimental results in graphic form, and for this different methods are available:

(1) The original, designated by me as the "spectral" mode of representation, which shows step by step the extent of diminution in the number of toxic doses effected by a definite small quantity of antitoxin, to wit, 1-200th of the immunity unit (which quantity I have called the "combining unit"), in the different zones of neutralization.

(2) The "linear" method, employed by Arrhenius, in which the quantities of antitoxin added are represented on the abscissa, and those of the active toxic doses present on the ordinates, is merely another form of graphic representation which has no advantage over the former. On the contrary this method chosen by Arrhenius, which is based upon estimation of the dissociation constants, is objectionable, because the graphic line corresponds only in its middle part with the values actually found, and toward the beginning and the end shows considerable deviations. Moreover its construction rests upon false premises.

INTERPRETATION OF THE PHENOMENA OF NEUTRALIZATION.

From the exhibited charts it is apparent that by employing the Lo mixture three different zones can be distinguished in the spectral mode of representation. In the first zone one combining unit of antitoxin removes on the average either one half or one entire lethal dose. As there is reason to assume that in the case of an ideal, absolutely pure toxin one combining unit removes exactly one lethal dose, the foregoing result indicates that in this zone (1) pure toxoid, or (2) pure toxin, or (3) a mixture of equal parts of toxin and toxoid, designated by me as "hemitoxin," is neutralized. The existence of hemitoxin is possible only in case the toxoid has precisely the same avidity as the toxin; from this it follows, as I have from the beginning assumed, and as Arrhenius has confirmed, that in the transformation of toxin into toxoid no change in avidity occurs.

Pure toxoid occurs often in the form of a prototoxoid, which possesses a stronger avidity for antitoxin than all other components of the toxin and thus manifests its presence by the fact that to the solution of toxin containing it a certain quantity of antitoxin can be added without lessening in the slightest degree the toxicity. The existence of prototoxoids has been recently denied by Arrhenius as a result of his recalculations of toxins studied by Madsen. His criticism, however, is certainly not applicable to a toxin which has been continuously tested by me, nor to another investigated with the utmost precision by Theobald Smith. Furthermore, prototoxoids have been positively demonstrated in the case of other poisons and analogous substances, ferments, agglutinins. Inasmuch as no change of avidity occurs in the formation of toxoids, the demonstration of prototoxoids points to the existence of a partial toxin of strong avidity — "prototoxin."

The second zone in the spectrum is of relatively short extent and is characterized by the fact that one combining unit removes only a fraction — one fifth to one tenth — of the lethal dose. This is the zone of decisive significance for Arrhenius's hypothesis.

The third zone I call the "toxone-zone." Toxone is one of the primary products of secretion of the diphtheria bacillus, possessing the same haptophore group as the toxin, but of far less avidity, and in the process of partial neutralization entering, therefore, last into combination with antitoxin. Its toxophore group differs from that of toxin, being incapable of producing acute effects, cutaneous necroses, death. Toxone is responsible for the diphtheric paralysis occurring between the fourteenth and thirtieth day in guinea pigs. Mixtures of toxin and antitoxin in which all the toxin-components are neutralized, while only the toxone remains unneutralized, produce in guinea pigs, without any local or general lesion, the phenomena of diphtheric paralysis.

PHYSICO-CHEMICAL OBJECTIONS AND THEIR REFUTATION.

Arrhenius by following a plan of experimentation introduced by Danysz has shown in the neutralization of ammonia by boric acid, that if 1 part of boric acid lessens the hemolytic effect of a given quantity of ammonia by 50%, the subsequent addition of 2, 3, 4 parts causes a further diminution of only 16.7, 8.3 and 5% respectively. It would be absurd to conclude from these experiments that ammonia is not a single substance but is composed of several constituents of various degrees of toxicity, of which those with the strongest chemical affinity are first neutralized. Arrhenius, therefore, concludes that inasmuch as the neutralization of tetanus-toxin proceeds in the same way it is superfluous to assume the existence of different components and sufficient to assume a single, uniform toxin with weak affinity for its antitoxin.

In reply I may state that at the outset of my investigations I was led to assume a strong affinity between toxin and antitoxin. This opinion in the first place was supported by the experiments of Dönitz and of Heymann who demonstrated the extraordinary rapidity with which diphtheria toxin introduced into the circulation combines with the toxophile receptors in the tissues, which must therefore possess a high degree of avidity. The protective and curative properties of the specific serum indicate that the antitoxins possess at least as strong avidity as the tissue-receptors. Furthermore, the therapeutic experiments of Dönitz on the animal body and of Madsen on red corpuscles prove that the toxin when once bound becomes so firmly fixed that the specific serum after a time fails to produce any curative effect. This indicates that after fixation of the toxin secondary, chemical modifications transform the primary, labile combination into a stable one.

The large interval between the values of Lo and L† seemed to oppose the acceptance of a strong affinity between toxin and antitoxin. Designate by D the amount of toxin which represents the difference between L† and Lo. From chemical examples it can be easily shown that with poisons of strong avidity the value of D must correspond exactly to one simple fatal dose, whereas with poisons of weak avidity D may be much larger. At last, however, I succeeded in finding a toxin in which D was precisely of the theoretic value 1. Thereby it was in principle shown that toxin and antitoxin unite with strong affinity, and the great variations (from 0–300%) in the value of D presented by different specimens of toxins could be referred to the presence of toxone. Not until I had found this toxin did I come forward with my theory, which I believe to have been framed with due care.

What then is the evidence advanced by Arrhenius in favor of the view that toxin and antitoxin are actually substances of weak avidity? The biological facts which speak against this view are either entirely ignored by the authors or set aside under arbitrary assumptions of an inexact investigation. The sole support is the tetanus-curve, concerning the value of which I shall speak presently. Experimental investigation should in the first instance have attacked this question.

The entire calculation and reasoning of the authors proceed from the assumption, adopted by so prominent an authority as Professor Nernst, that neutralization of toxin and antitoxin is an interchangeable or reversible reaction. If it can be shown that the process is not a reversible one, then the basis is withdrawn from all the reckonings and deductions of Arrhenius. Unaccountably Arrhenius has presented no experiments upon this point, but Dr. v. Dungern has taken up the subject by applying the procedure used by Danysz for other purposes of successive additions of toxin to antitoxin. By way of example the processes in the production of acetic ether may be selected. Mix 1 equivalent of alcohol with ⅓ equivalent of acetic acid, allow the mixture to stand for a length of time and then add ⅔ equivalent of acetic acid (mixture A); at the same time mix simultaneously 1 equivalent of acetic acid and 1 equivalent of alcohol (mixture B). It is self-evident that the state of equilibrium, when two thirds of the added substance have been transformed, is reached sooner in A than in B. If the two mixtures are examined at a time when equilibrium is established in B, both liquids have of course the same composition. Before this point is reached, however, the transformation is advanced further in A than in B. If we assume in this example that acetic acid corresponds to toxin, alcohol to antitoxin and the ester to the neutral antitoxin-toxin mixture, it is readily seen that by the addition of acetic acid in two periods the mixture A is deprived of toxicity sooner than is the control B with simultaneous addition.

Von Dungern's experiments were made with a diphtheria poison of which the Lo dose was 0.6 ccm. and the L† dose 0.78. To 1 immunity unit of antitoxin he added first one third of the Lo dose — 0.2 toxin, and after twenty-eight hours determined the fatal dose of toxin. It was found that this mixture after the addition to it of only 0.39 ccm. of toxin was fatal to guinea pigs. Thus in accord with Danysz's previous observations, it was demonstrated that an increase of toxicity appeared when the toxin was added in two portions, since under these conditions the L† dose was reduced from 0.78 (in the case of simultaneous mixture) to 0.59. (If antitoxin was added in successive fractional portions of the immunity unit to the toxin solution the L† dose of course remained 0.78.) From his experiments von Dungern concludes:

(1) The union of diphtheria toxin and antitoxin does not proceed in accordance with the ammonia-boric-acid scheme.

(2) The observed phenomena of combination are explicable only upon the assumption of a complex constitution of the diphtheria poison.

(3) The facts find their best explanation in the action of a toxone and epitoxonoid. Epitoxonoid is present in considerable quantity in toxic broth. The immunizing action of apparently completely neutralized toxins can thereby be explained.

(4) Constituents of the diphtheria poison with weak affinity after combining with antitoxin may eventually become so firmly bound that this union can be only incompletely broken by toxin of stronger avidity. The strength of the combination is of significance for the action of antitoxin.

(Demonstration of the relations by schematic charts.)

At my suggestion Dr. Sachs has undertaken the same experiments with tetanolysin. In one experiment 0.02 ccm. tetanus serum served as the immunity unit. This neutralized completely 0.3 ccm. (Lo dose) of a definite toxin solution, while a mixture containing 0.7 (L†) toxin effected complete hemolysis. By adding to the serum the Lo dose (0.3) and again after twenty hours adding more toxin the L† dose was found to be reduced to 0.45. It is consequently demonstrated also for tetanolysin, which is the axis of Arrhenius' investigations, that the premises underlying his entire calculations and construction of graphic curves are inapplicable. Hereafter the linear representation of the calculated values of toxins, as demanded by Arrhenius for all cases, must be rejected wherever the phenomenon of Danysz is present.

Also from the purely physico-chemical point of view, as I am authoritatively informed, the conceptions of the authors present numerous vulnerable points which will soon be exposed by competent authority. I confine myself here to the presentation of certain opposing arguments of a biological nature, and begin with tetanus.

Tetanolysin, on account of its ready decomposability and the difficulty of conducting hemolitic experiments with absolute accuracy, is ill adapted for precise determinations of the kind under consideration. Neutralization experiments with this substance encounter on the one side the Scylla of incomplete union and on the other the Charybdis of destruction of the poison.

It is incomprehensible that the calculations should be based on the assumption that the toxin consists of molecules of poison with uniform composition. On the contrary, it is highly probable that specific toxoids are present, which, of course, must enter into the calculation as well as the toxins. The amount of toxoid was, however, unknown and indeed at present is indeterminable. To escape from this dilemma Arrhenius, as he has kindly informed me, assumes that the production of toxoid results from an equable weakening of all the toxic molecules. Thus the toxic solution would not be a mixture of dissimilar components. This assumption must be rejected on general grounds (examples) and is disproven by the demonstration of prototoxoids, as well as by the investigations of Tarobi on ricin-toxoids.

It is likewise incomprehensible that Arrhenius should leave entirely out of account the tetanophilic receptors of the red corpuscles and make no calculation of the amount of poison fixed by the corpuscles. These anchor no small part of the simple dissolving dosis, which is the dominating factor in the experimental procedure.

Even upon the assumption of a toxin of uniform composition the calculation of the state of equilibrium must, therefore, take into consideration three factors capable of combining with antitoxin — toxin, toxoid and receptor.

To meet these objections, which might easily be augmented, Arrhenius has scarcely anything of a positive character to offer save the tetanolysin table No. 19, which corresponds fully with the theoretical calculation. It must here be emphasized that in contrast with a large number of unsuccessful results this favorable one was secured but once. I have lately become very skeptical as regards the demonstrative value of this curve, inasmuch as both of Madsen's diphtheria curves, which on account of their correspondence with the calculation was published as a complete refutation of my conceptions, have since been shown to be erroneous. I communicated by letter to Professor Arrhenius that the numbers underlying these curves were in entire harmony with my conceptions, and I was astonished to be informed that a recalculation occasioned by my criticism showed that a part of the figures underlying the construction of the curves were wrong. Without my letter Madsen's curves would probably have stood for decades as an argument against my views. At any rate these considerations demonstrate that from erroneous figures, curves in entire harmony with the theory may be constructed.

At present there are no satisfactorily demonstrated facts which warrant the application of Arrhenius' conceptions to the diphtheria toxin. It may be mentioned that an extensive investigation by Morgenroth has positively overthrown Madsen's assumption that the toxone-effect is merely an expression of the action of toxin modified by antitoxin.

The large variations of the dissociation-constants, as determined by the calculation, necessitate the assumption of an extraordinary multiplicity of poisons, whereas my conceptions are satisfied by the acceptance of a much smaller number.

It is unnecessary to have recourse to further arguments when I mention in conclusion that my views furnish a far better explanation than those of Arrhenius of numerous facts in the domain of toxins, such as the process of spontaneous enfeeblement of toxins, their passage into a state of stability, the reduction of the toxone zone, the capacity of completely neutralized poisons to produce antitoxin (Park) by means of von Durgern's epitoxonoids. I am convinced that what I have established through long years of work concerning the nature of the diphtheria poison will withstand the Northern storm and with the course of time prove itself to be real and vital.

LECTURE III.

CYTOTOXINS AND CYTOXIC IMMUNITY.

Abstract.

Development of our knowledge concerning bacteriolysins and cytolysins. Bacteriolysins (Pfeiffer's phenomenon in cholera) and their mode of action. Agglutinins (Gruber, Durham and Widal). Cytolysins (Belfanti, Metchnikoff, Bordet, von Dungern, Landsteiner).

Importance of this domain constantly increasing. Literature of recent years almost too great to master.

The hemolysins especially suitable for theoretical studies; they made it possible for me to explain quite simply the manner in which these substances arise, and the mode of their action.

MODE OF ACTION OF AMBOCEPTORS AND COMPLEMENTS.

The interpretation of Pfeiffer, Metchnikoff and Bordet's demonstration of the participation of two substances in producing bacteriolysis was made clear by means of absorption experiments and experiments at low temperatures. Establishment of the amboceptor theory. The amboceptor (the specific product of immunization) unites by means of one haptophore group with the receptor of the susceptible cell, and by means of another haptophore group with the complement. The complement has one chemical group suitable for union with the amboceptor, and another, the zymotoxic complex — which causes the injury to the cells. Bordet's opposition; his theory of "sensitization."

For the theory of direct union (between cell-receptor, amboceptor and complement) the following evidence may be given:

(1) Demonstration of well-established chemical analogues.